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UNSUPERVISED LEARNING

Task– Discover hidden patterns or structures in the data to potentially reveal patterns like diabetes risk levels or other health conditions.



Experience— Suppose we have a dataset of individuals with the following columns:

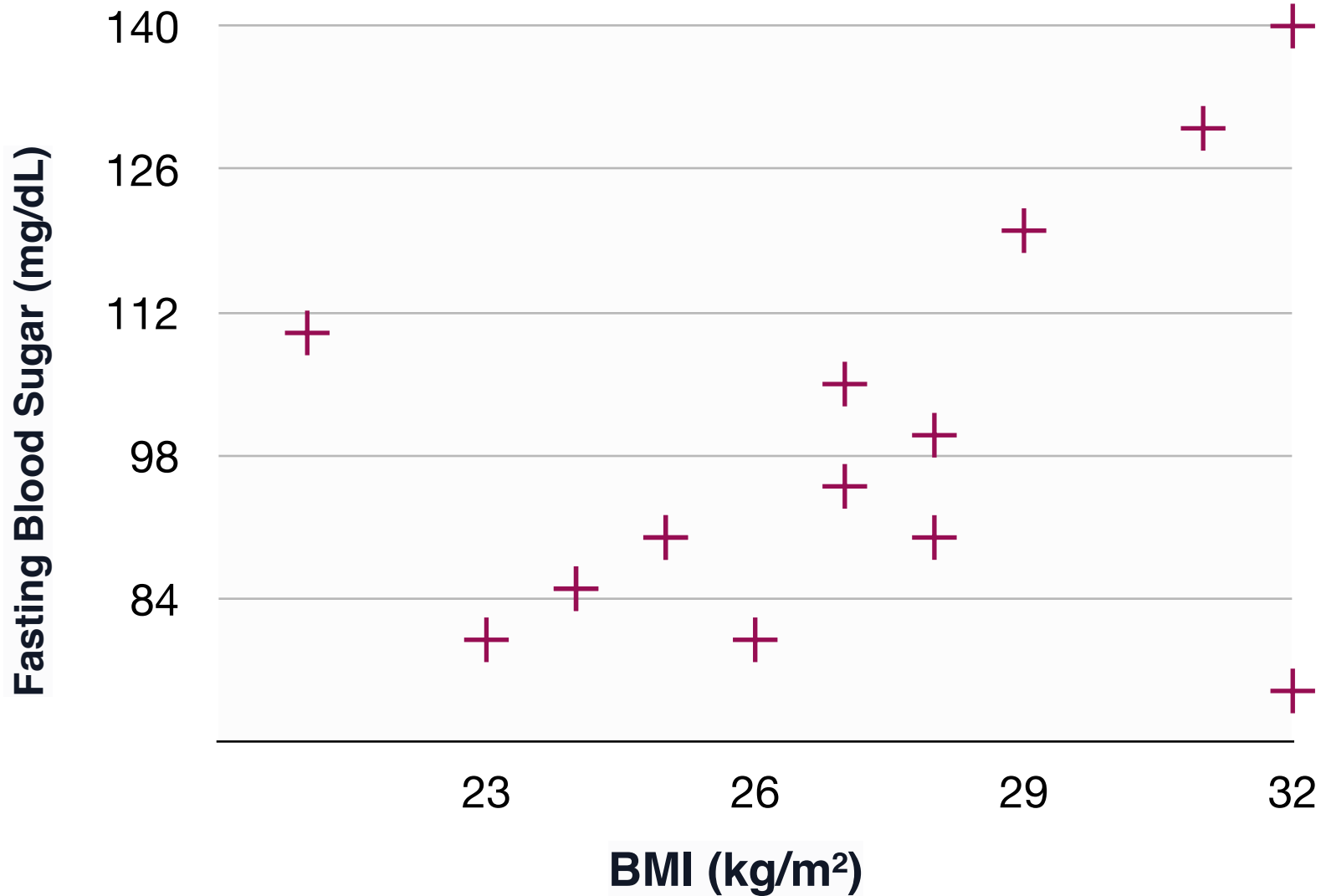
- ▶ x_1 : Fasting blood sugar level

- ▶ x_2 : Body Mass Index (*BMI*)

- ▶ Unlike supervised learning, unsupervised learning does not use labeled outcomes like diabetes development variable.

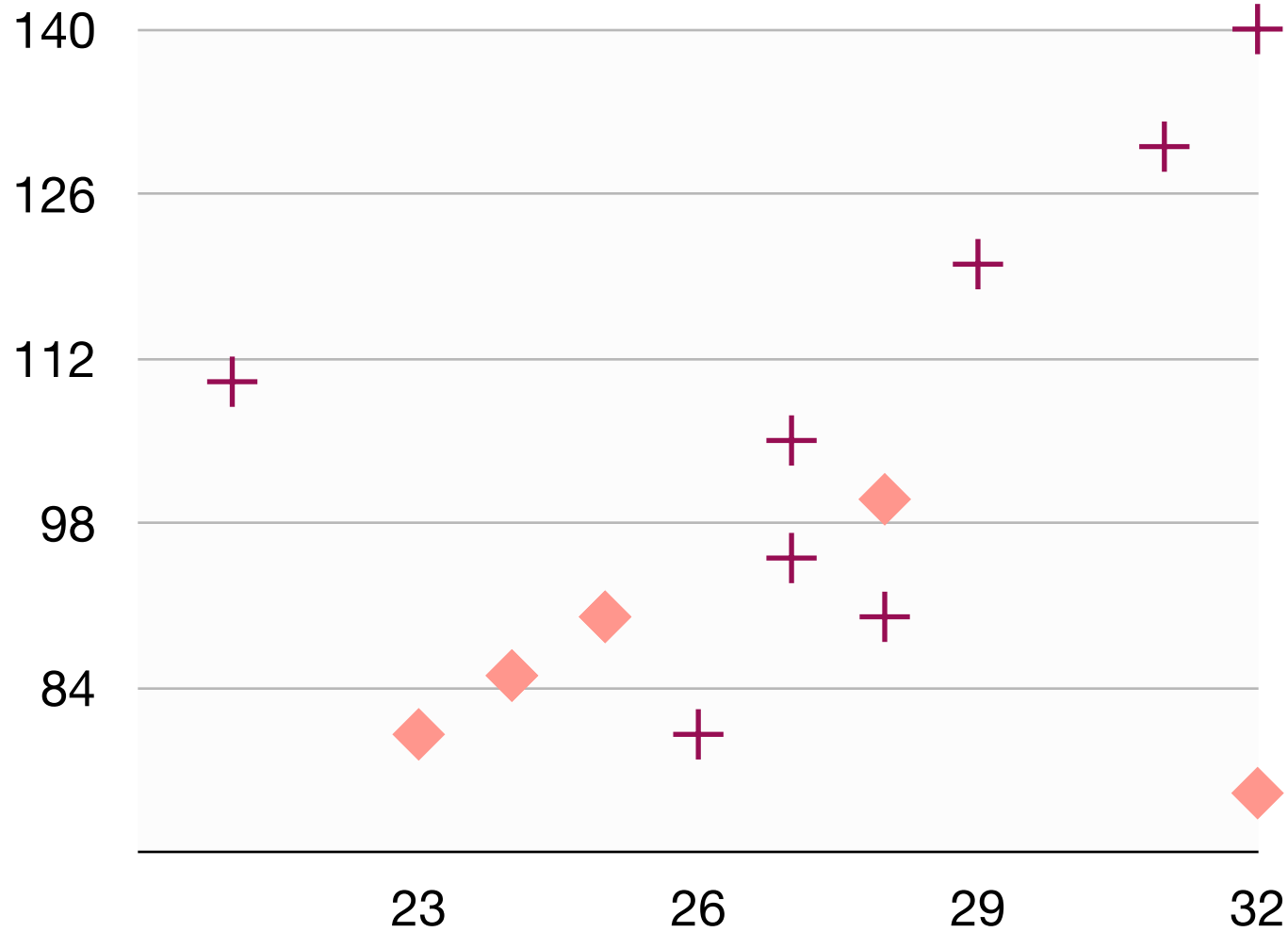


Clustering



Classification

Fasting Blood Sugar (mg/dL)



Label/Class

- + Diabetes=1
- ◆ Diabetes=0

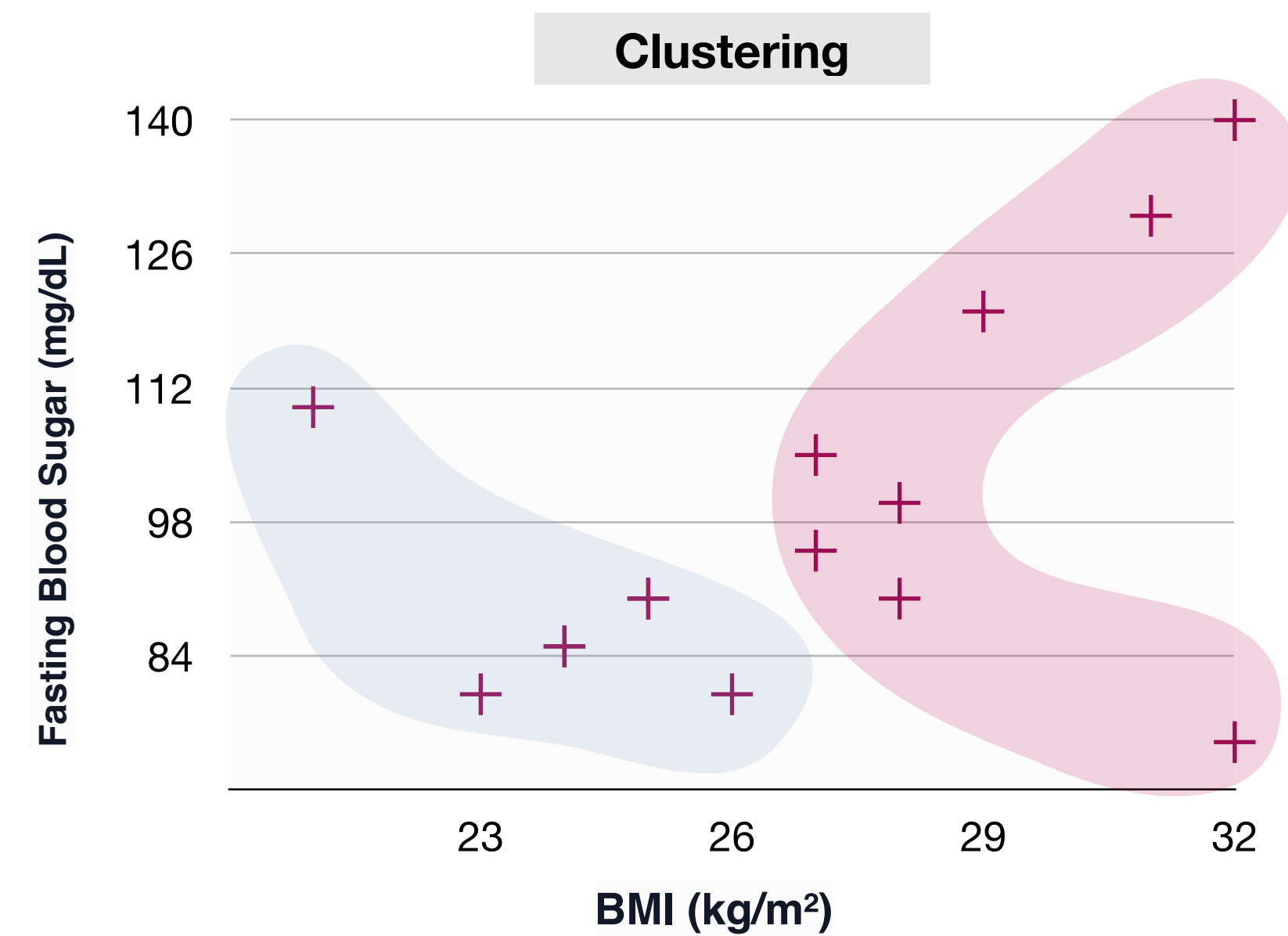
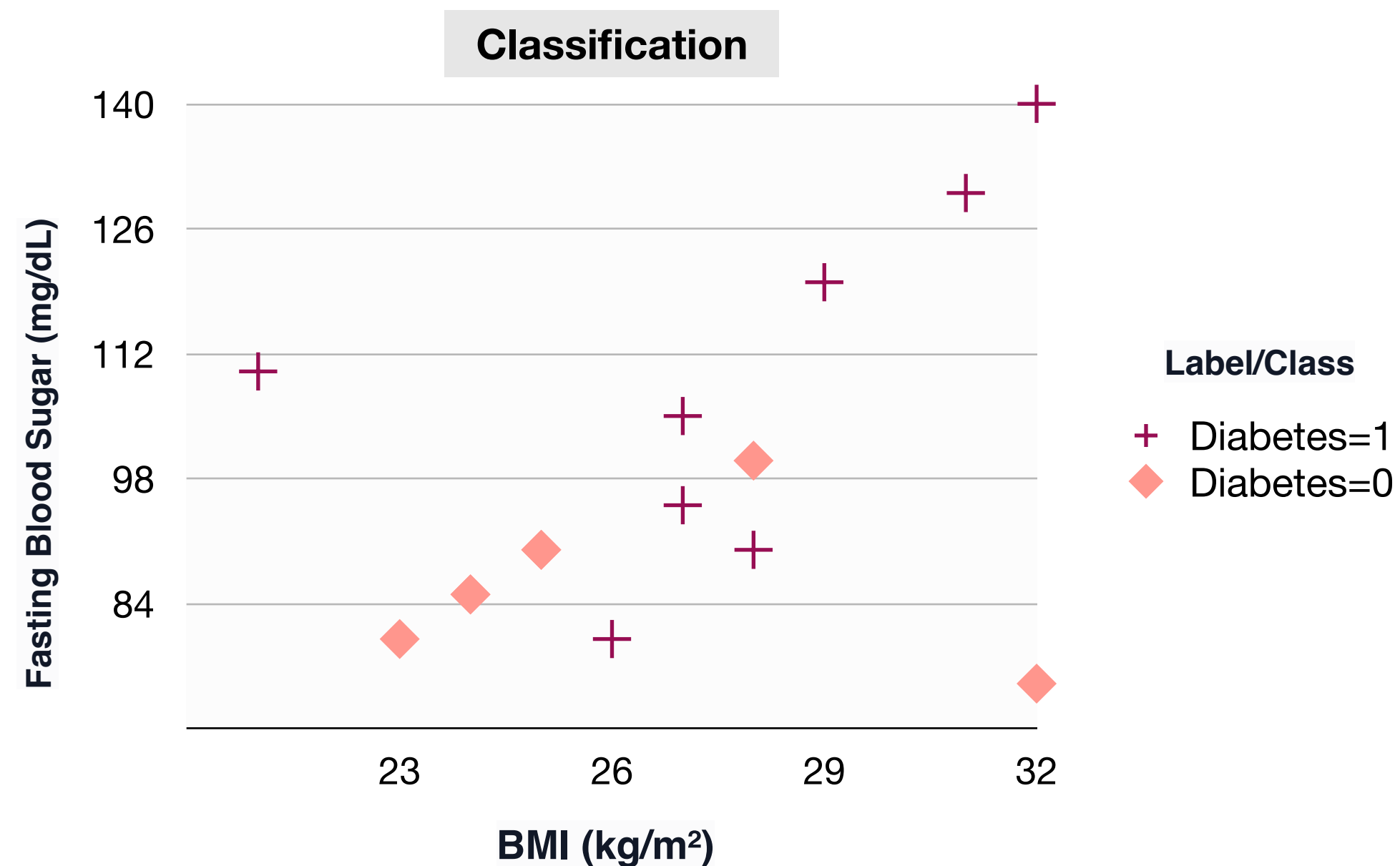
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Word Embeddings



Word embeddings are unsupervised learning models trained on extensive text data, devoid of labels or annotations.

They are a way to convert words into numerical vectors so that similar words are placed close to each other in a mathematical space.

